

# MUHAMMAD UMAIR ASHRAF

**Nationality:** Pakistani **Date of birth:** 15/10/2002 **Gender:** Male
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**Home:** *MI City Depalpur, District Okara, Punjab, Pakistan*
GitHub LinkedIn Portfolio

## PERSONAL PROFILE

Final-year Software Engineering student with a strong academic record and applied research experience in computer vision and AI-based behavioural modelling. Final year project (Vision-Code) involved designing, evaluating and benchmarking a multi-signal AI system combining gaze estimation, head pose estimation, face recognition, and a novel per-person calibration method against published methods from the literature. Seeking to pursue a Master's degree in Computer Science to build on this research foundation.

## EDUCATION AND TRAINING

**BS Software Engineering** [ 2022-2026 ]
**Institution:** University of the Punjab, Lahore, Pakistan | **CGPA:** 3.02 / 4.00
Final Year Project: Vision-Code Online learning platform integrating an AI-based real-time exam proctoring system.

**Higher Secondary School Certificate (HSSC)** [ 2019 – 2021 ]
**Major Subjects:** Physics, Chemistry, Biology | **Institution:** Punjab College, Okara (BISE Sahiwal) | **Result:** 1083/1100 (98.5%), Grade A+

**Secondary School Certificate (SSC)** [ 2017 – 2019 ]
**Major Subjects:** Physics, Chemistry, Biology, Mathematics | **Institution:** A.R. High School of Science, Depalpur, Okara (BISE Sahiwal) | **Result:** 1063/1100 (96.6%), Grade A+

## RESEARCH EXPERIENCE — FINAL YEAR PROJECT

**Vision-Code: A Real-Time AI Exam Proctoring System Using Gaze Tracking, Head Pose Estimation, Face Recognition, and Audio Analysis** [ Apr 2026 ]
*Final Year Project — Department of Software Engineering, PUCIT*

Designed a rule-based, per-candidate-calibrated attention-monitoring model and independently benchmarked it against seven published gaze/attention-tracking methods from the literature (including CNN-based, transformer-based, and CNN-LSTM hybrid models), re-evaluating all of them on an identical 2,995-frame dataset to ensure a fair, like-for-like comparison rather than citing each paper's own reported numbers.

**Methodology**

- Head pose estimation: mapped six facial landmarks (nose tip, eye corners, mouth corners, chin) to a 3-D model via OpenCV's solvePnP to compute yaw/pitch, using per-person calibrated baselines rather than fixed population thresholds.
- Gaze estimation: tracked iris position from 478 3-D facial landmarks (MediaPipe Face Mesh) to compute normalized gaze deviation from a calibrated baseline.
- Face recognition / identity verification: implemented and compared two backends a lightweight OpenCV Haar Cascade histogram model and an ArcFace/InsightFace deep embedding model (512-D) evaluating accuracy/latency trade-offs (97.5% vs 90.3% verification accuracy).
- Designed a rule-based multi-signal attention classifier fusing eye-aspect-ratio (blink/eye-closure), yaw/pitch deviation, gaze deviation, and audio RMS thresholding into a single attentive/inattentive decision per frame.
- Ran a controlled ablation study isolating the effect of per-person calibration: precision dropped from 93.7% to 80.3% when calibration was removed, confirming it as the primary driver of low false-positive rates.

**Results**

- Achieved 93.7% precision, 94.1% recall, 93.9% F1-score, and 91.6% accuracy on a live evaluation session (2,995 frames), outperforming the best re-evaluated literature baseline (a transformer-based gaze/head model) by 10.6% in precision without requiring a GPU.
- Authored a full technical paper documenting the related-work review, system design, experimental methodology, and comparative evaluation (currently unpublished / in preparation for submission).

***Tech:*** *Python, OpenCV, MediaPipe, ArcFace / InsightFace, computer vision, statistical evaluation (precision/recall/F1/confusion matrix)*

## WORK EXPERIENCE

**Associate Software Engineer:** Devkiosk, Lahore — May 2026 to Present
**Associate Software Engineer (Intern):** Sortup, Lahore — Mar 2025 to Jun 2025
Full-stack development roles building web applications and RESTful APIs.

## CERTIFICATIONS

- Microsoft .NET Blazor — Udemy Certificate, 2025
- The Ultimate React Course 2025 — Udemy Certificate, 2026

## TECHNICAL SKILLS

**Frontend:** React.js, Redux, JavaScript, TypeScript, HTML5, CSS3, Tailwind CSS, Bootstrap
**Backend:** Node.js, Express.js, MongoDB (MERN), .NET Core, Django, FastAPI, REST API Development
**Languages:** JavaScript, TypeScript, C#, Python, C++
**Databases:** MongoDB, PostgreSQL, Microsoft SQL Server
**AI / CV:** OpenCV, MediaPipe, ArcFace / InsightFace, Computer Vision pipelines
**Tools & Cloud:** Git, GitHub, Postman, Entity Framework, Prisma, Supabase, Microsoft Azure, Vercel, Render, CI/CD

## LANGUAGE SKILLS

Urdu

English

## REFERENCES

**Dr. Muhammad Farooq | Assistant Professor, Department of Information Technology, PUCIT**
*Final Year Project Supervisor*
**Email:** mfarooq@pucit.edu.pk

**Mr. Fareed-Ul-Hassan Baig | Assistant Professor, Department of Software Engineering, PUCIT**
Course Instructor: Programming Fundamentals and Object Oriented Programming
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